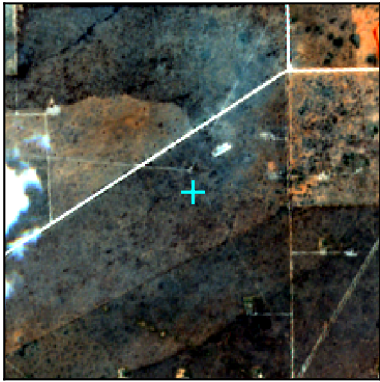
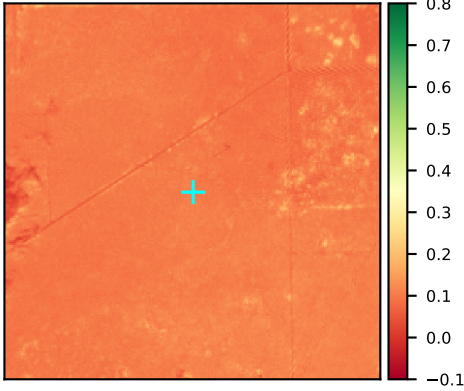


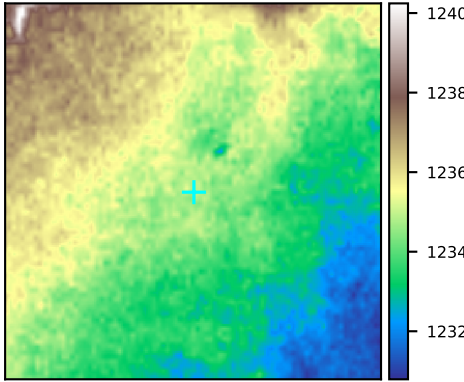
Sentinel-2 RGB
2022-07-09 — context only, NOT the model input



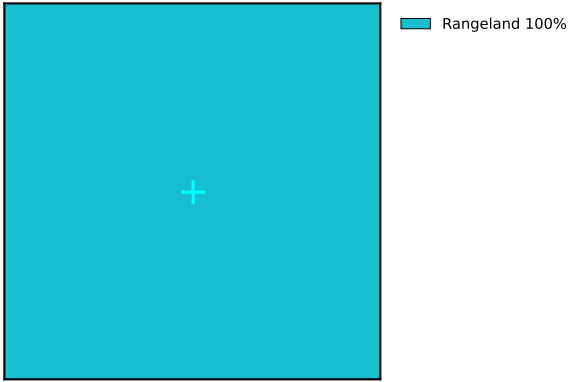
NDVI
ñ=0.139



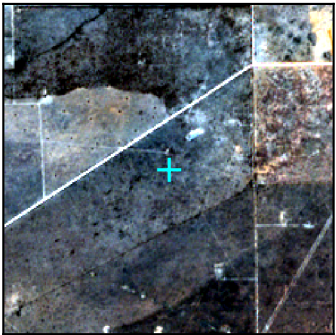
DEM [m]
relief 9 m, ñ=0.001



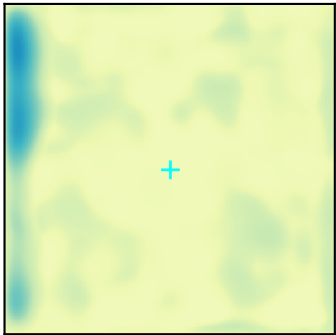
Land cover (ESRI v2 → TerraMind idx)



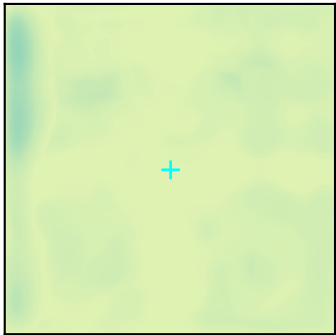
DJF 2022-01-15
MODEL INPUT: S2 2022-01-15, lag 0 d, cloud 0.00



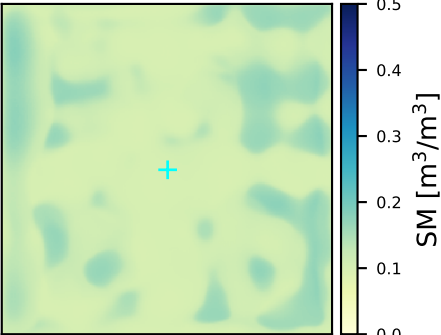
0-10 cm mean 0.078
σ 0.0408 · ñ 0.524 · Δ 0.188



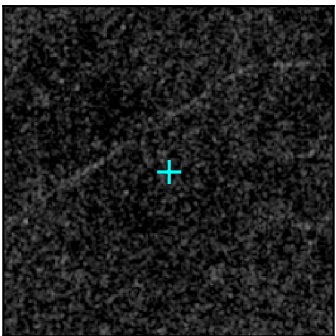
10-30 cm mean 0.098
σ 0.0132 · ñ 0.135 · Δ 0.060



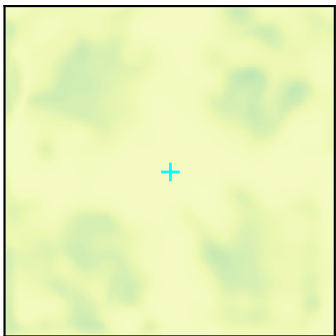
30-100 cm mean 0.120
σ 0.0218 · ñ 0.182 · Δ 0.070



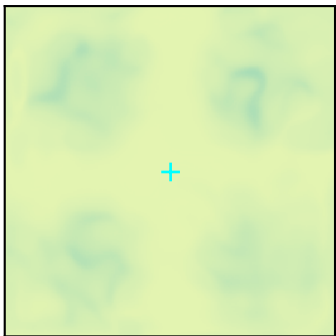
MAM 2022-04-15
MODEL INPUT: S1_DESC 2022-04-08, lag 7 d



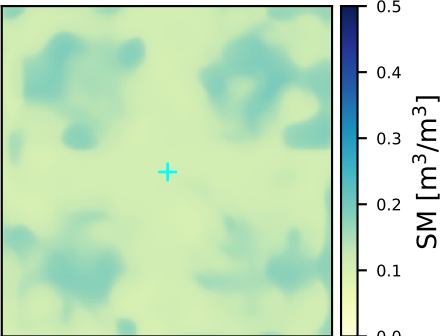
0-10 cm mean 0.059
σ 0.0212 · ñ 0.357 · Δ 0.072



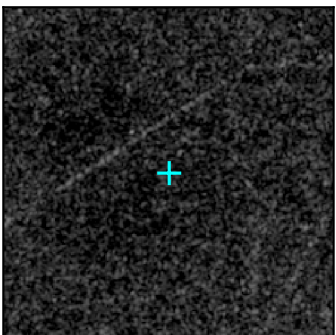
10-30 cm mean 0.096
σ 0.0160 · ñ 0.167 · Δ 0.055



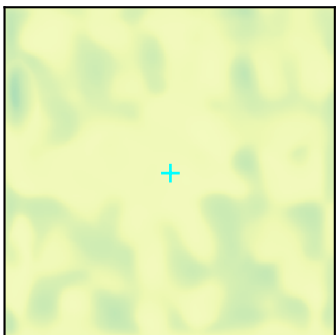
30-100 cm mean 0.127
σ 0.0237 · ñ 0.186 · Δ 0.079



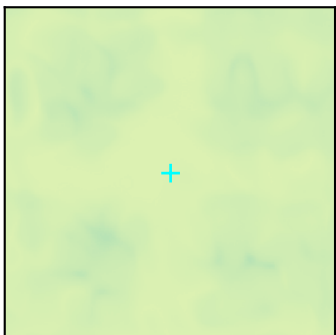
JJA 2022-07-15
MODEL INPUT: S1_DESC 2022-07-13, lag 2 d



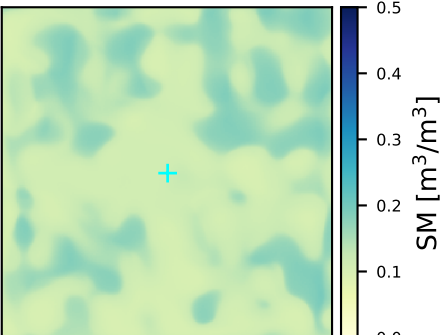
0-10 cm mean 0.070
σ 0.0220 · ñ 0.317 · Δ 0.073



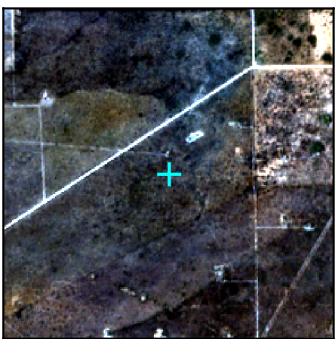
10-30 cm mean 0.104
σ 0.0093 · ñ 0.089 · Δ 0.034



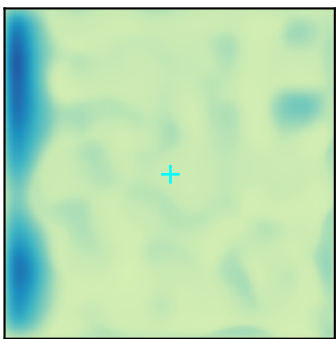
30-100 cm mean 0.130
σ 0.0248 · ñ 0.190 · Δ 0.082



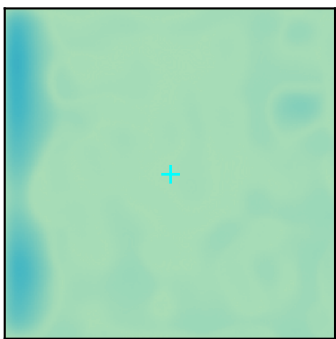
SON 2022-10-15
MODEL INPUT: S2 2022-10-12, lag 3 d, cloud 0.00



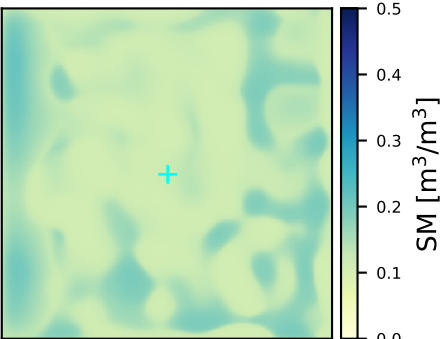
0-10 cm mean 0.139
σ 0.0461 · ñ 0.332 · Δ 0.206



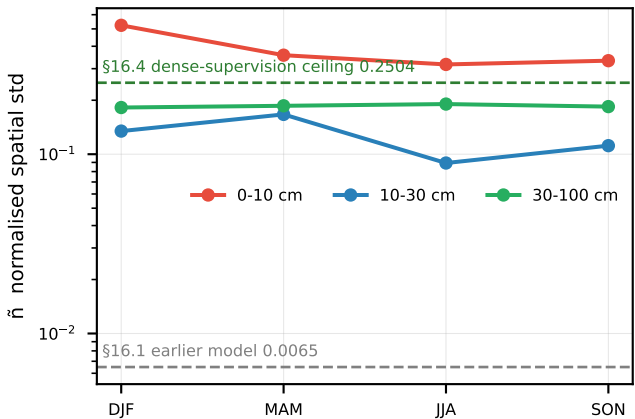
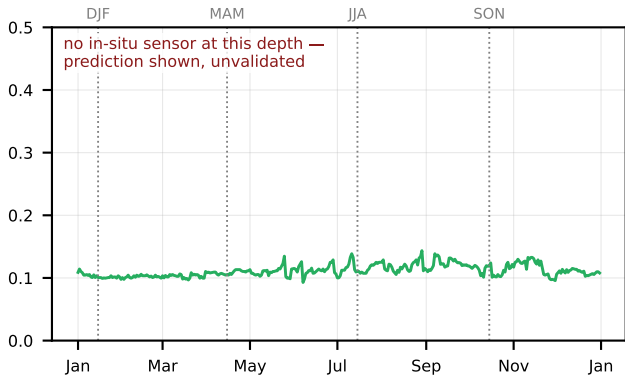
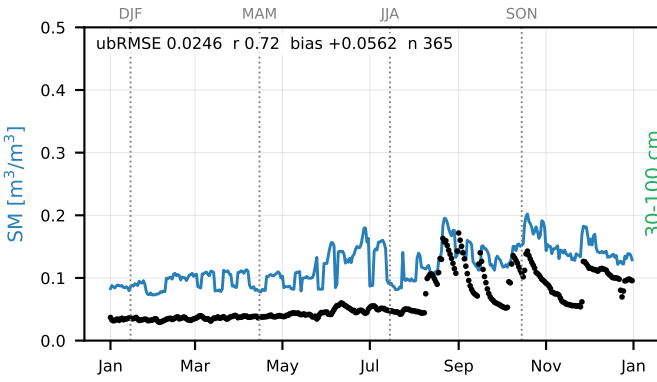
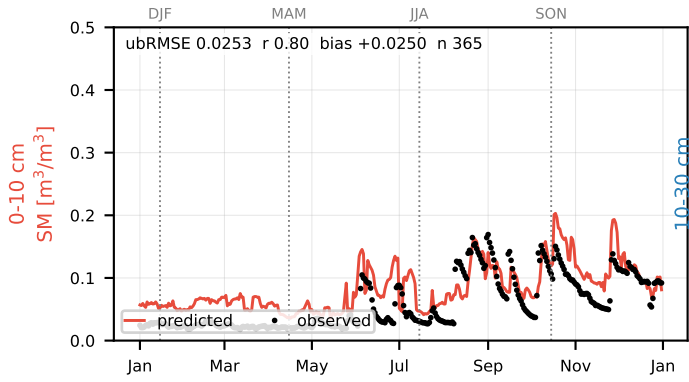
10-30 cm mean 0.161
σ 0.0180 · ñ 0.112 · Δ 0.079



30-100 cm mean 0.135
σ 0.0248 · ñ 0.184 · Δ 0.083



model at the station pixel (112,112), 2022 — season markers tie each map above to this curve



SYMBOLS
σ spatial standard deviation of the predicted map [m³/m³]
σ time temporal std of the station-pixel prediction, this year
ñ normalised spatial std = σ / [mean of map] [-]
Δ spread of the map, 98th minus 2nd percentile [m³/m³]
r Pearson correlation coefficient [-]
14² 14x14 token grid = model bottleneck, ≈160 m pixels
var kept at 14² share of map variance surviving block-averaging to that grid
r(season-season) similarity of the spatial pattern between seasons: 1 = one fixed pattern, 0 = unrelated
anchor, lag the satellite acquisition the model actually consumed, and its age in days
ubRMSE unbiased RMSE = √(MSE - bias²) [m³/m³]
SM volumetric soil moisture [m³/m³]

SPATIAL SUMMARY
0-10 cm σ 0.0325 σ time 0.0365 σ/σ time 0.89
r(DEM,14²) +0.21 r(NDVI,14²) -0.07
var kept at 14² 79% r(season-season) +0.34
10-30 cm σ 0.0141 σ time 0.0307 σ/σ time 0.46
r(DEM,14²) +0.17 r(NDVI,14²) -0.10
var kept at 14² 84% r(season-season) +0.37
30-100 cm σ 0.0238 σ time nan σ/σ time nan
r(DEM,14²) +0.02 r(NDVI,14²) -0.05
var kept at 14² 74% r(season-season) +0.38
cross-depth r(0-10, 30-100) +0.37 — architectural (star residual, model.py:276), not learned depth structure.
HOW TO READ
r(DEM/NDVI) ≈ 0 → the structure is not the terrain or the vegetation of this footprint
r(season-season) → 1 → one pattern, mean-shifted
var kept at 14² ≈ 100% → no detail below ≈160 m
σ/σ_time ≈ 1 → across 2.2 km the model varies as much as it does across the whole year